

LOGO1



UZ Logo

Revision

Rev01

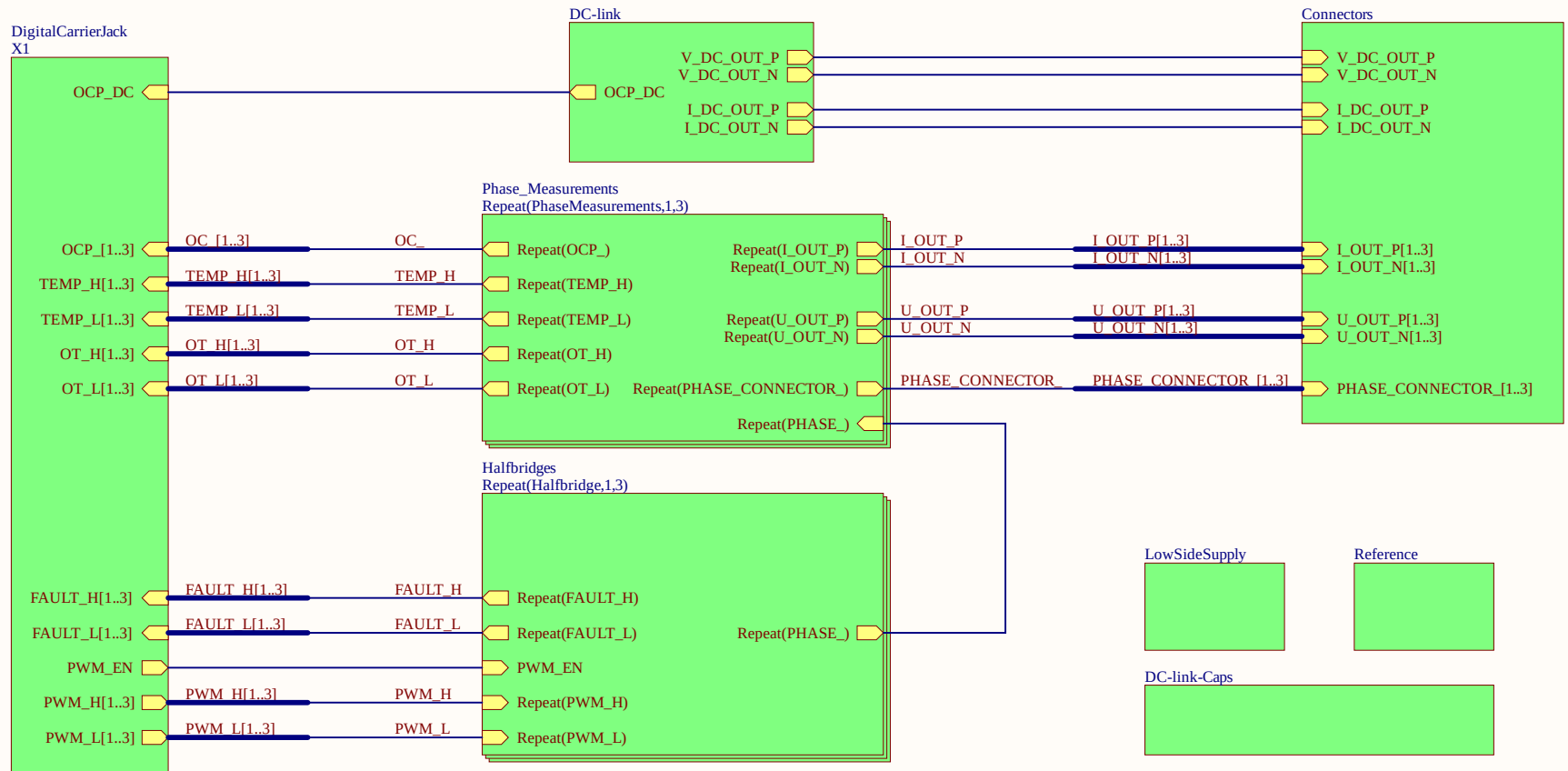
Serial

Serial #

INFO1

Project
ProjectRevision
AuthorParam
ProjectDate

Design Information



Title TopSheet.SchDoc

Revision: Rev01

Design Engineer: S. Anzer

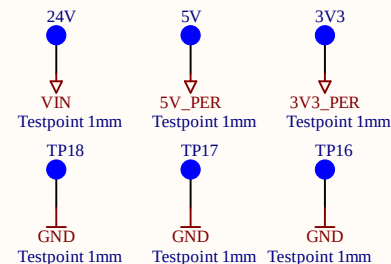
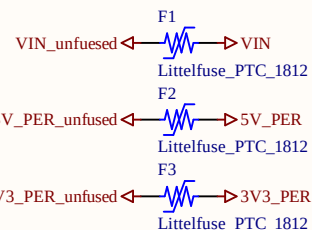
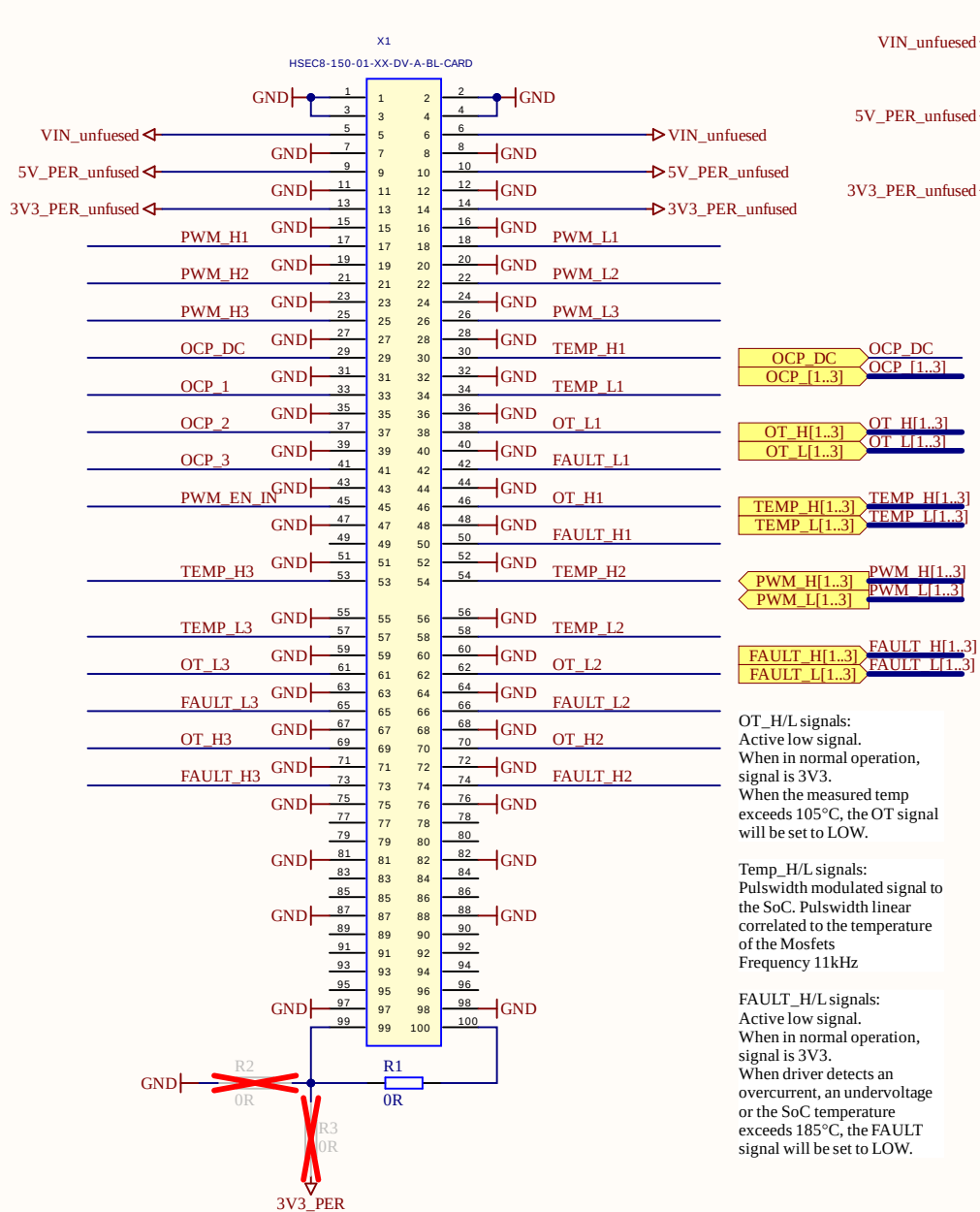
Project: GanInverter.PrjPCB

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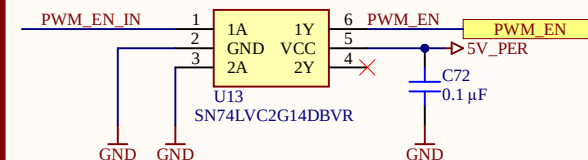
Date: 29.12.2022

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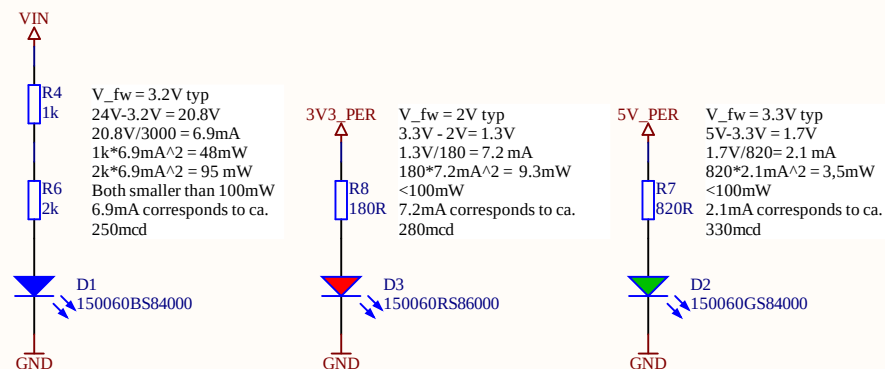




Inverting Schmitt Trigger



Status LEDs for UZ-supply rails



Title DigitalCarrierJack.SchDoc

Revision: Rev01

Design Engineer: S. Anzer

Project: GanInverter.PrjPCB

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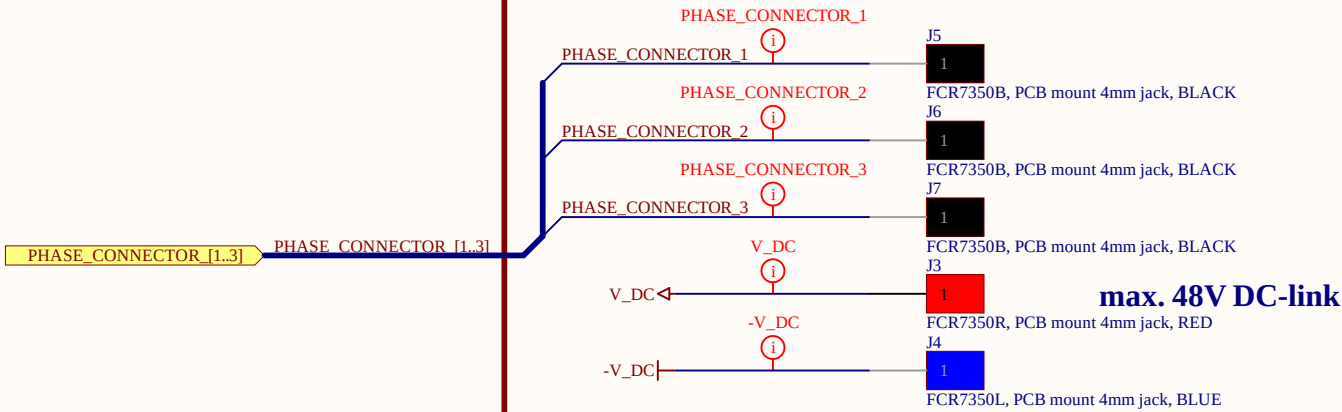
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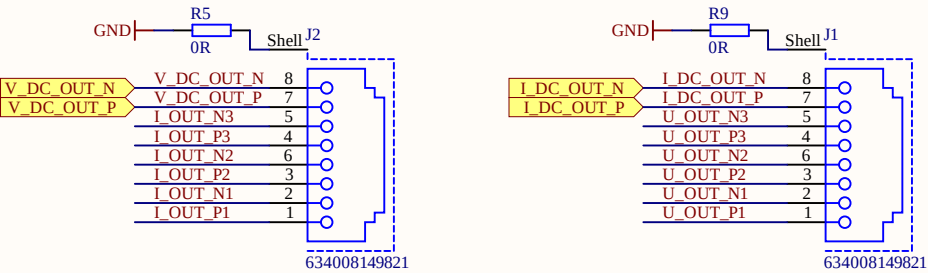
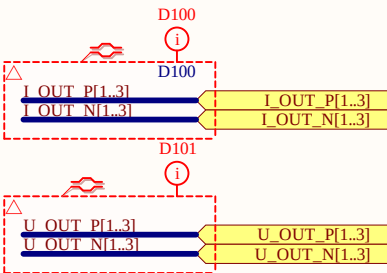
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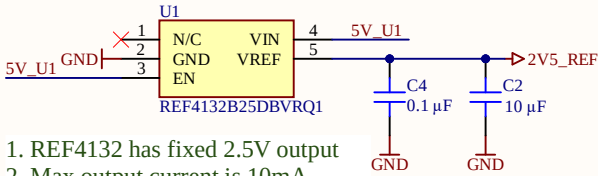
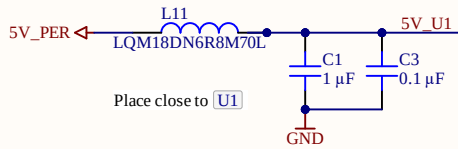
High Power - Banana Connectors



Ethernet Jacks



2.5V Reference Voltage

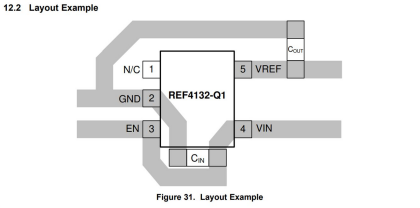


- 1. REF4132 has fixed 2.5V output
- 2. Max output current is 10mA

12.1 Layout Guidelines

Figure 31 illustrates an example of a PCB layout for a data acquisition system using the REF4132-Q1. Some key considerations are:

- Connect low-ESR, 0.1-μF ceramic bypass capacitors at V_{IN} , V_{REF} of the REF4132-Q1.
- Decouple other active devices in the system per the device specifications.
- Using a solid ground plane helps distribute heat and reduces electromagnetic interference (EMI) noise pickup.
- Place the external components as close to the device as possible. This configuration prevents parasitic errors (such as the Seebeck effect) from occurring.
- Do not run sensitive analog traces in parallel with digital traces. Avoid crossing digital and analog traces if possible, and only make perpendicular crossings when absolutely necessary.

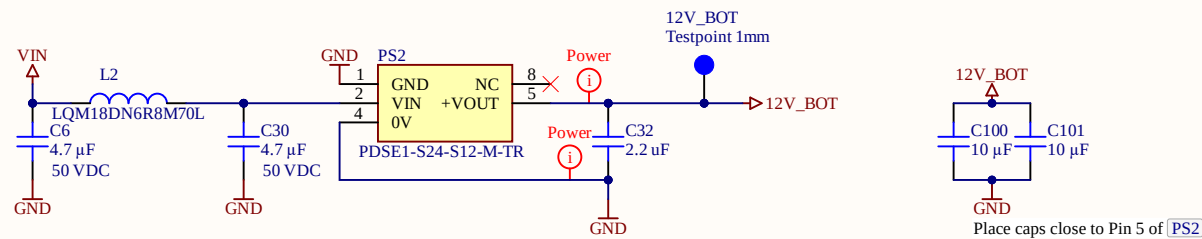


10.2.1 Design Requirements

A detailed design procedure is described based on a design example. For this design example, use the parameters listed in Table 2 as the input parameters.

Table 2. Design Example Parameters	
DESIGN PARAMETER	VALUE
Input voltage V_{IN}	5 V
Output voltage V_{OUT}	2.5 V
REF4132-Q1 input capacitor	1 μF
REF4132-Q1 output capacitor	10 μF

12V Bottom-Side Voltage Supply



- 1. PSDE1 has an input range from 21.6V - 26.1V.
- 2. Fixed 12V output voltage. Max. 84mA / 1W.

Estimated Power consumption Bottom Side:

Gatedriver **U16**:
VDD: 12V
IDD: max. 3.5mA
Power consumption: 42mW

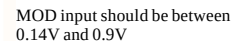
Fet **Q2**:
QG: max. 16nC
UDRV: 5.8V
fSW: 100kHz
Power consumption: 98.2mW

Total consumption: 3 * (42mW + 98.2mW) = 404mW

4



C

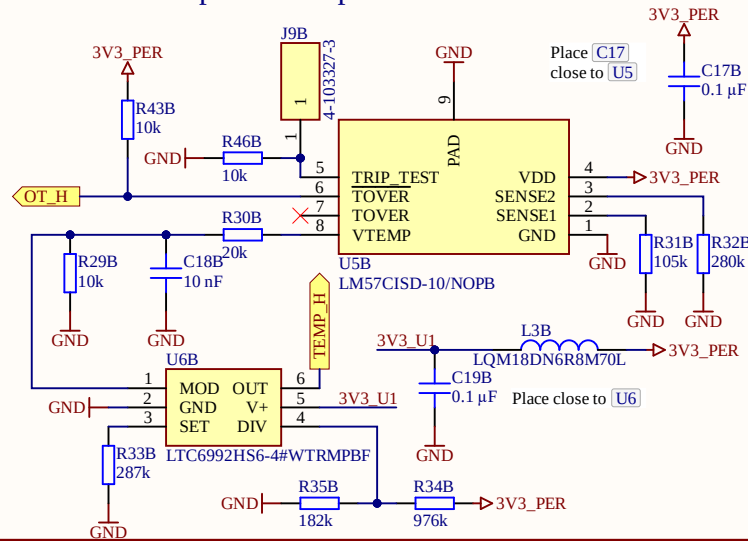


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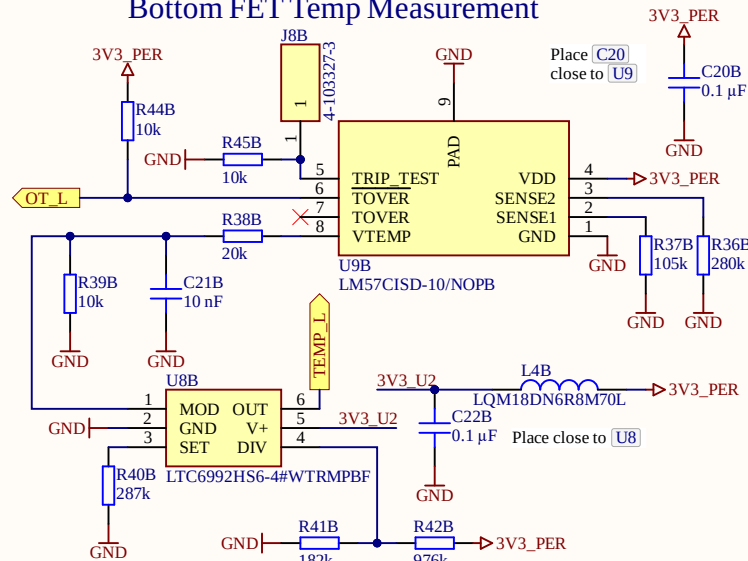
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Top FET Temp Measurement

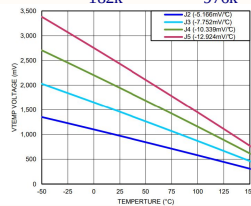


Bottom FET Temp Measurement



R_SENSE1 = 105k
R_SENSE2 = 280k is Gain
->J4 ->Leads to:
1.11V@105°C
2.00V@20°C

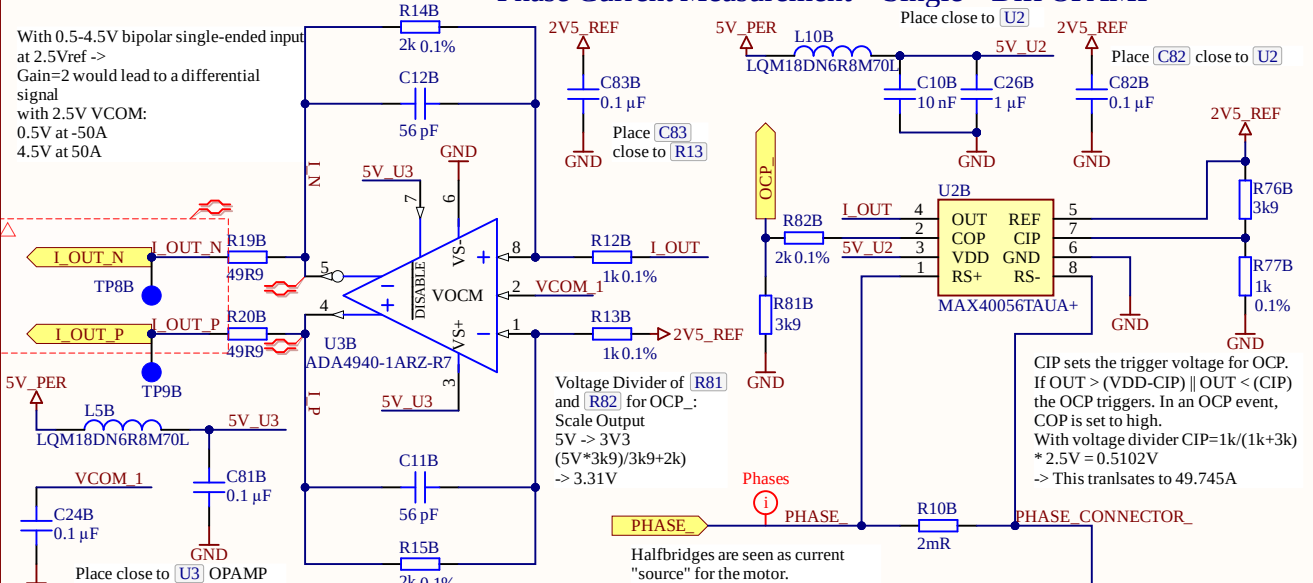
/TOVER trips at 105°C



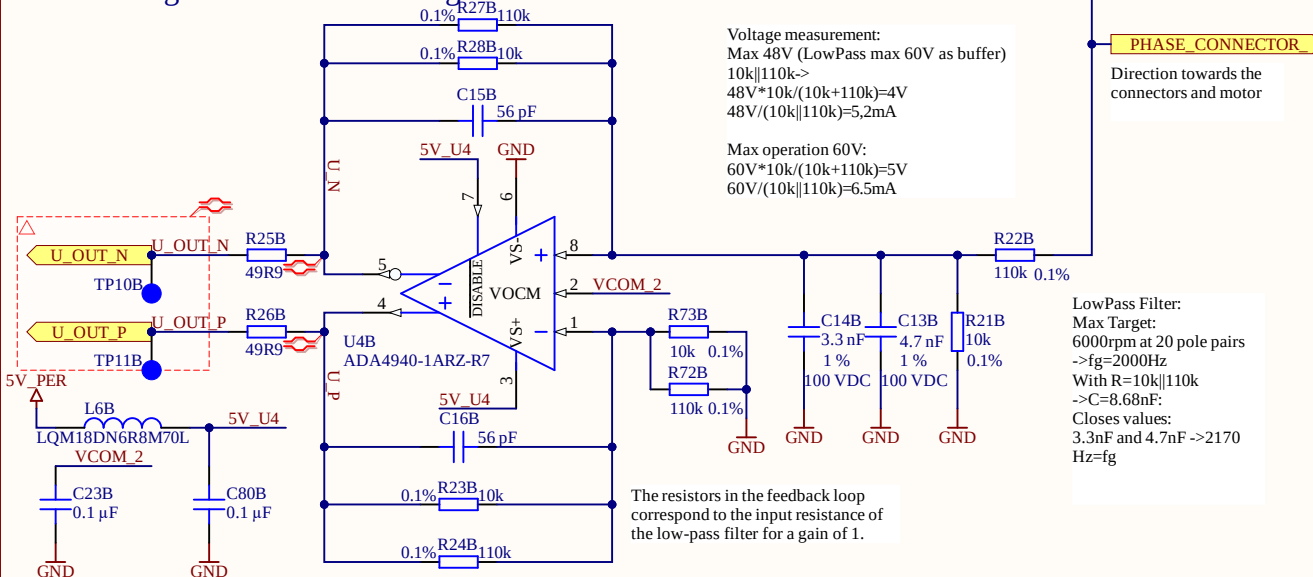
VTEMP has voltage divider with 20k/10k:
MOD input voltages->
 $1.11V * 10/(10+20) = 0.37V @ 105^{\circ}C$
 $2.00V * 10/(10+20) = 0.67V @ 20^{\circ}C$

MOD input should be between
0.14V and 0.9V

Phase Current Measurement + Single->Diff OPAMP



Phase Voltage Measurement + Single->Diff OPAMP

Title Phase_Measurements.SchDoc

Revision: Rev01	Design Engineer: S. Anzer
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Project: GanInverter.PrjPCB

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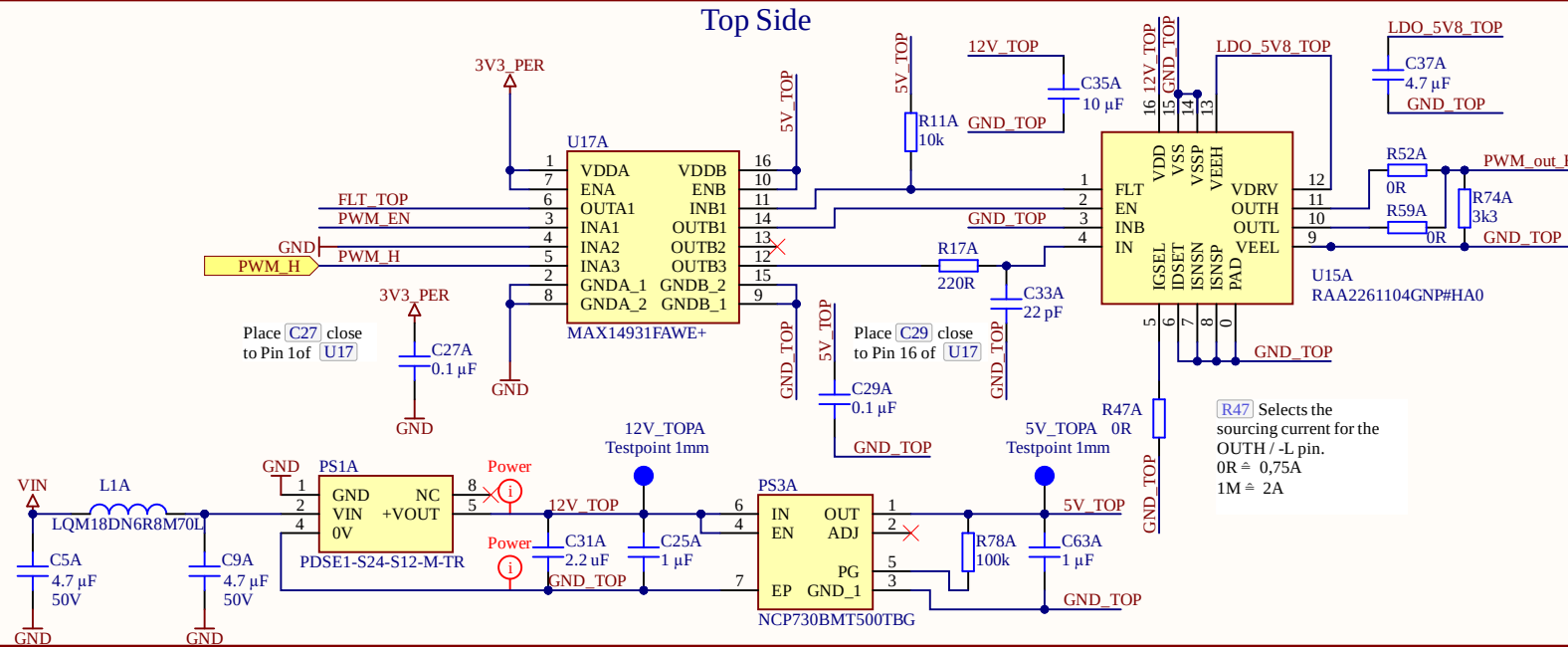
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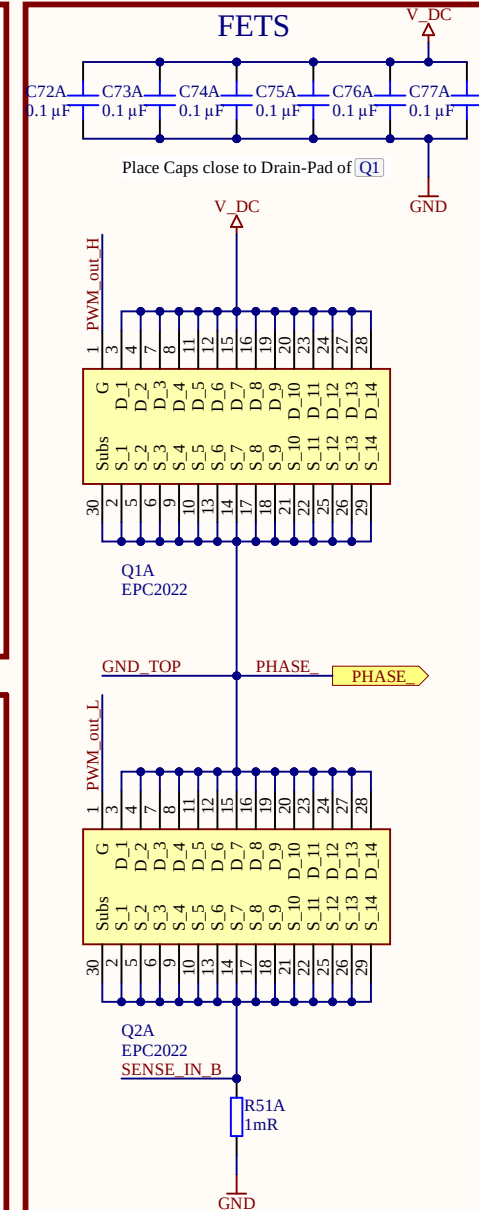
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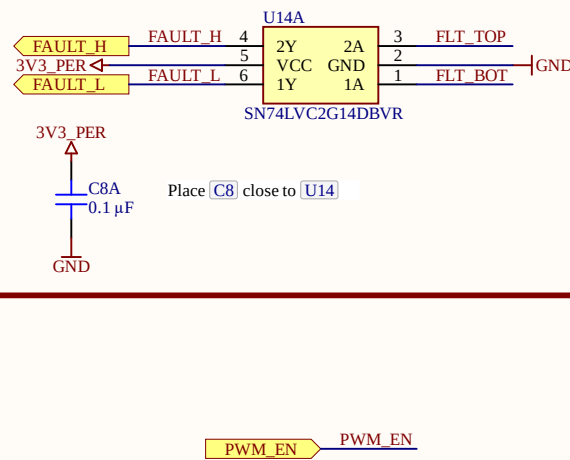
Top Side



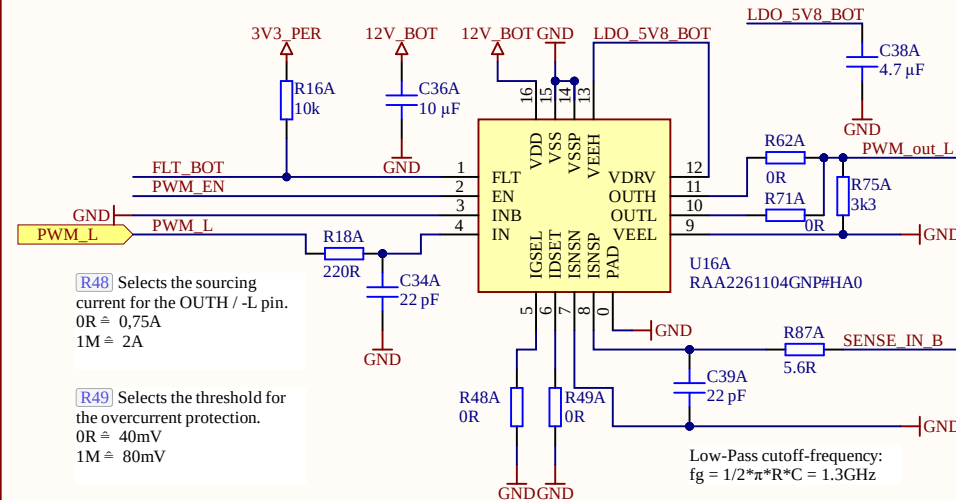
FETS



Inverting Schmitt Trigger



Bottom Side



Title Halfbridges.SchDoc

Revision: Rev01

Design Engineer: S. Anzer

Project: GanInverter.PrjPCB

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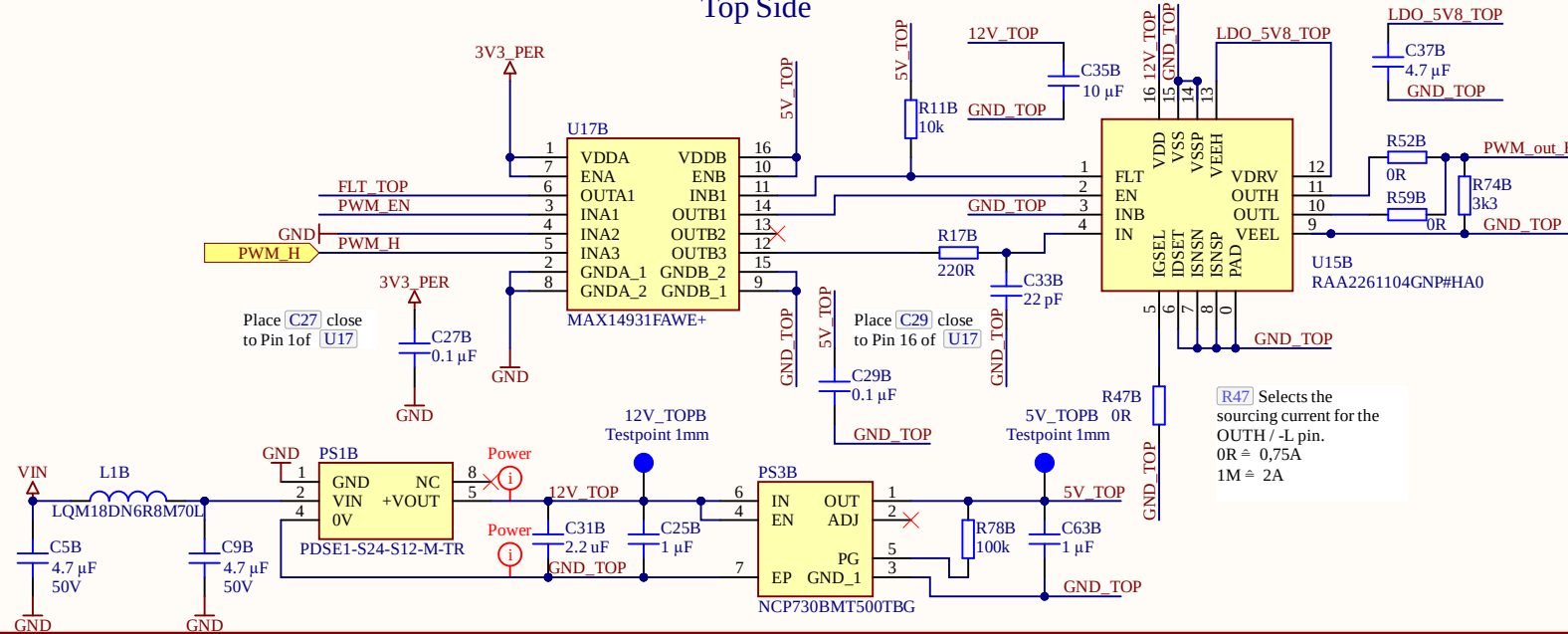
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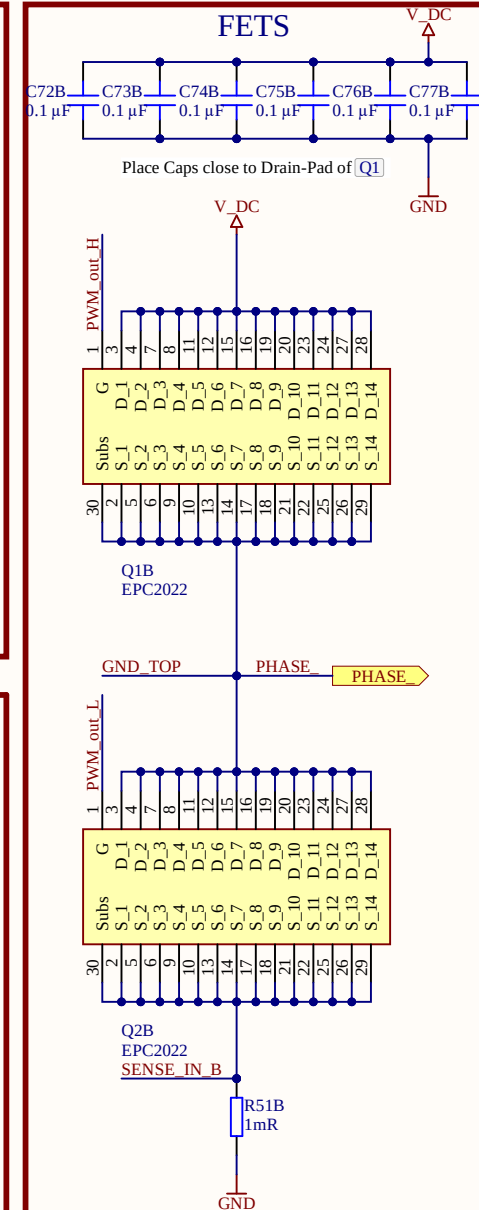
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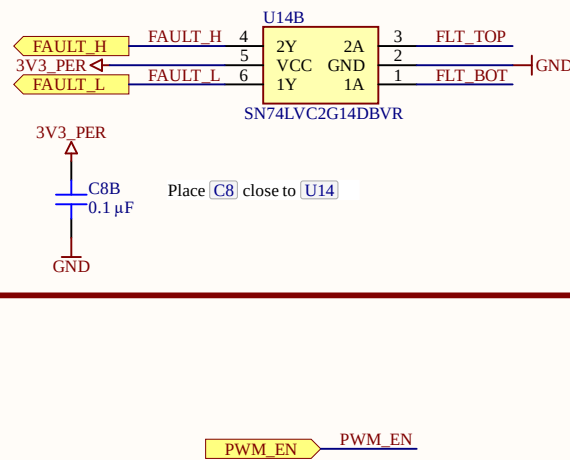
Top Side



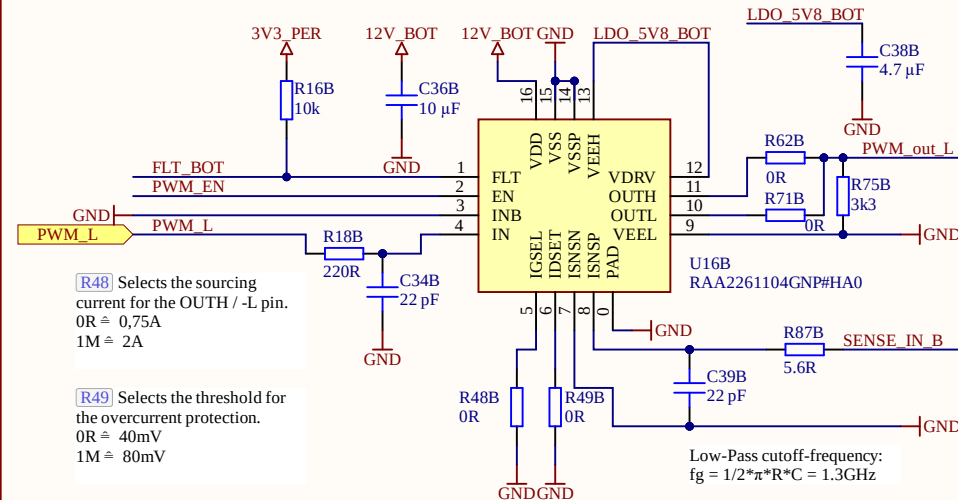
FETS



Inverting Schmitt Trigger



Bottom Side



Ground connection of ISNSN pin needs to be connected directly to [R51] GND pad.

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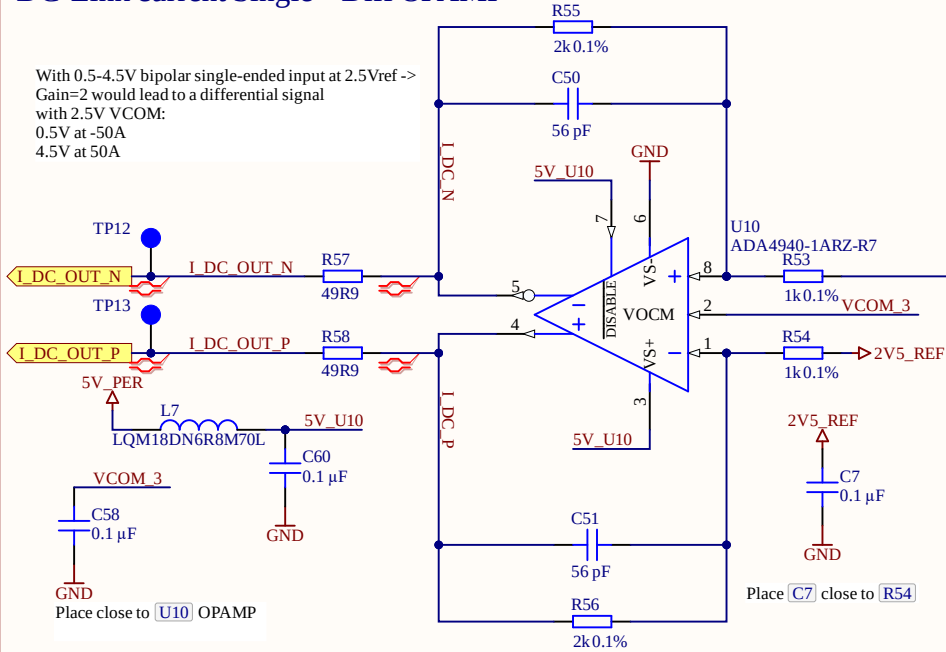
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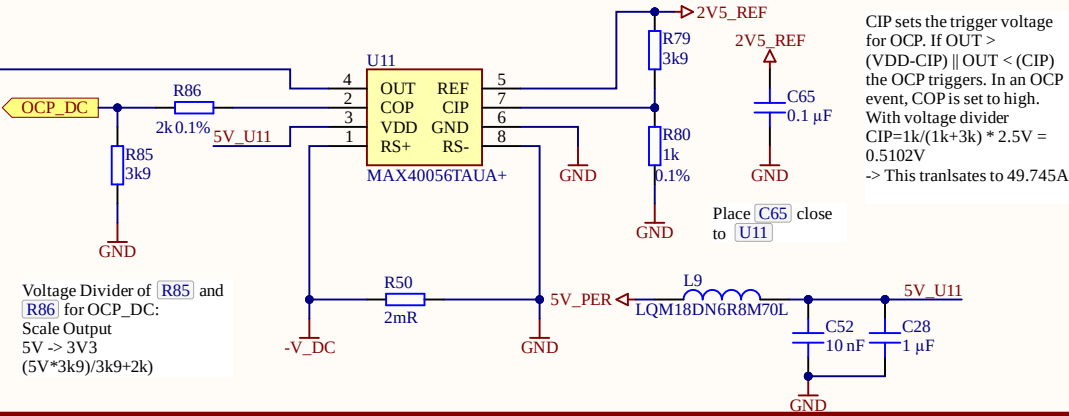


DC-Link current Single->Diff OPAMP

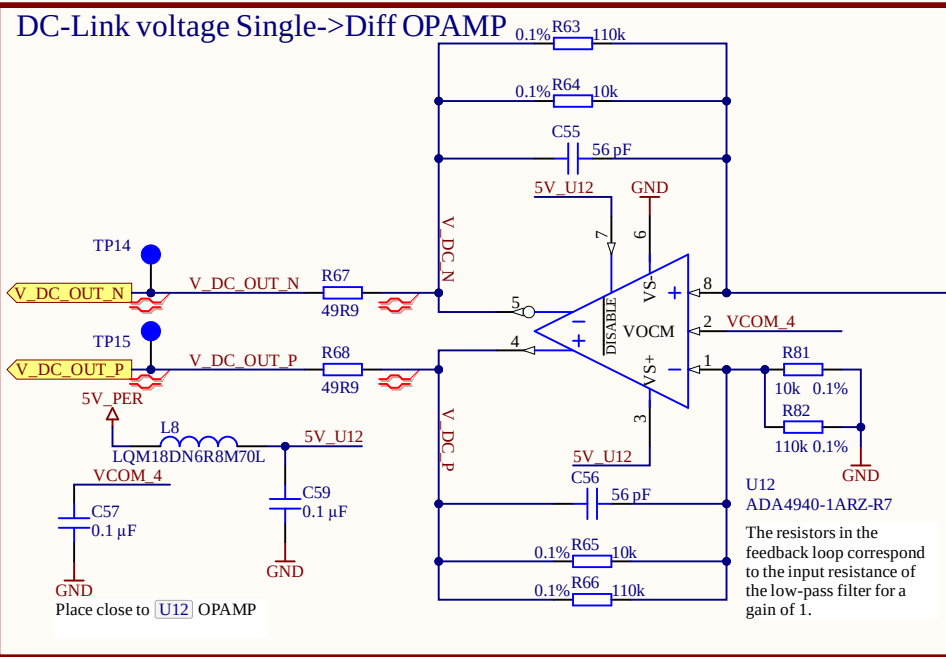
With 0.5-4.5V bipolar single-ended input at 2.5Vref ->
Gain=2 would lead to a differential signal
with 2.5V VCOM:
0.5V at -50A
4.5V at 50A



DC-Link current measurement



DC-Link voltage Single->Diff OPAMP

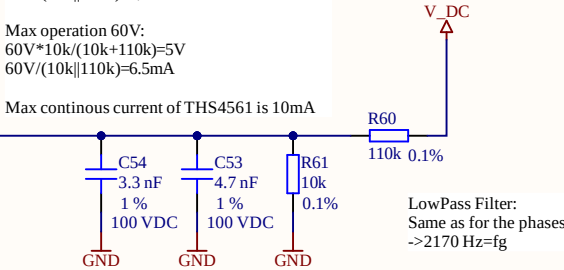


DC-Link Voltage lowpass

Voltage measurement:
Max 48V (LowPass max 60V as buffer)
 $10k \parallel 110k \rightarrow$
 $48V * 10k / (10k + 110k) = 4V$
 $48V / (10k \parallel 110k) = 5.2mA$

Max operation 60V:
 $60V * 10k / (10k + 110k) = 5V$
 $60V / (10k \parallel 110k) = 6.5mA$

Max continous current of THS4561 is 10mA



Title DC-link.SchDoc		 UltraZohm www.ultrazohm.com
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DC-link capacitance:
 $(10 \cdot 47 \mu\text{F}) + (12 \cdot 10 \mu\text{F}) = 590 \mu\text{F}$

